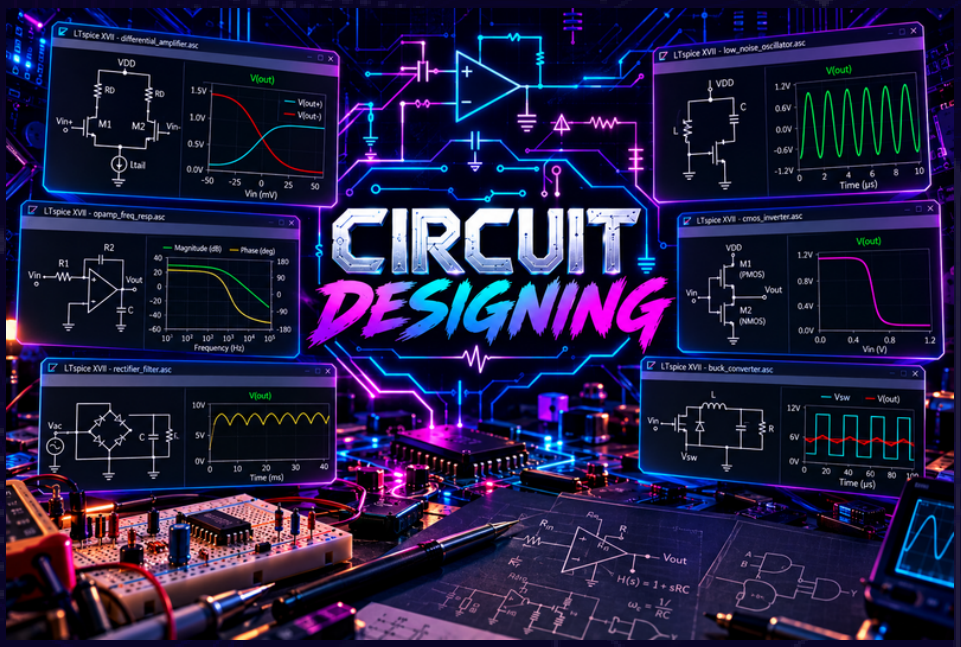
NIT ARUNACHAL PRADESH
ADDOVEDI
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CIRCUIT DESIGNING



1. Introduction

Circuit Designing is a two-round Analog Electronics and Circuit Design competition organised by the ECE Society as part of Addovedi, the annual technical fest of NIT Arunachal Pradesh. The event tests participants' understanding of analog electronics and circuit analysis, along with their ability to design, simulate, analyze and defend circuits using LTspice. The competition progresses from conceptual fundamentals in Round 1 – Analog Electronics Quiz to applied, time-bound circuit design and simulation with Viva in Round 2.

2. Event Details

- **Event Name:** Circuit Designing
- **Organising Body:** ECE Society, NIT Arunachal Pradesh
- **Fest:** Addovedi
- **Date, Time and Venue:** To be announced by the organisers to registered participants.
- **Event Coordinator:** Anirudh Mishra (contact no. 9140463181)

3. Eligibility and Registration

1. **Eligibility:** The event is open to undergraduate students from recognised institutions.
2. **Team Size:** Each team may consist of a maximum of 2 participants.
3. **Registration:** Each participant may register only once; duplicate registrations will be disregarded, and the registration fee is ₹50 per member. Registration is available on Unstop.
4. **Student ID:** Every participant must carry a valid student ID card issued by their institution and produce it on demand. In case of any dispute or discrepancy, the decision of the organisers and judges shall be final and binding.
5. The organisers reserve the right to modify any or all of these rules. Any changes will be communicated to registered participants.

4. Event Structure

4.1 Round 1: Qualifying Round

Mode: MCQ-Based Screening Round

Round 1 will test the fundamental knowledge of analog electronics and circuit analysis of the participants. The quiz will cover:

- Circuit Analysis
- Semiconductor Fundamentals
- Diode Circuits
- Basic Analog Electronics

Format

- **Total Questions:** 25 MCQs
- **Regular Questions:** 20
- **★-Marked Questions:** 5
- **Duration:** 30 Minutes
- **Round 1 Marks:** 25
- **Negative Marking:** No
- The 5 ★-marked questions are used only for tie-breaking.

4.2 Round 2: Competitive Round

Mode: LTspice-Based Circuit Design Round

Round 2 will test the participants' ability to design, simulate, analyze, and explain analog circuits using LTspice. The circuit design may cover:

- RC / RL / RLC Circuits
- Low-Pass & High-Pass Filters
- Diode Circuits
- Rectifier, Clipper & Clamper Circuits
- Other Basic Analog Circuits

Format

- **Total Marks:** 75
- **Software:** LTspice
- **Circuit Design:** Participants will be given a circuit design problem with specified requirements.
- **Simulation:** Participants must design and simulate the required circuit within the allotted time.
- **Viva:** Participants may be required to demonstrate and explain their circuit.

Evaluation

Round 2 carries 75 marks, including Circuit Design and Viva. The detailed evaluation criteria and distribution of marks will be decided by the judges/Organizing Committee based on the given problem and competition requirements.

5. Software and Tools

- The permitted software is LTspice, used for circuit design and simulation.
- Participants must bring their own laptops, with LTspice installed and working beforehand. The organisers will not provide computers.
- Laptops must be fully charged, and participants should bring their own chargers. The organisers are not responsible for any hardware, software or battery failure.
- Participants may not install or use any other software, extension or tool unless permitted by the organisers.

6. General Rules

- Participants must report at the venue at the time announced by the organisers. Late entry may be refused.
- Internet access, mobile phones, storage devices, notes and any other aids are not permitted during the rounds unless specifically allowed by the organisers.
- The use of artificial intelligence tools, code generators or any external help is strictly prohibited.
- All work submitted must be the participant's own. Copying, sharing of solutions or any form of plagiarism will result in disqualification.
- Participants must not interfere with the laptops, files or work of other participants.
- Participants must follow the instructions of the coordinators and judges at all times and maintain discipline and decorum at the venue.
- Any form of misconduct, unfair means or violation of these rules may lead to penalty or disqualification at the discretion of the judges.

7. Tie-Breaking Rules

Tie-breaking will be applied only after the complete final score out of 100 has been calculated.

First Tie-Breaker

The number of correctly answered ★-marked questions in Round 1 will be compared. The participant with more correct ★ answers will rank higher.

Second Tie-Breaker

If the number of correct ★ answers is also the same, the Round 2 score will be compared. The participant with the higher Round 2 score will rank higher.

Final Tie-Breaker

If the tie still remains, the Organizing Committee may conduct an additional tie-breaker.

8. Evaluation and Results

- Evaluation will be carried out by the judges appointed by the organisers.
- The final score will be calculated based on the combined scores of Round 1 and Round 2.
- In the event of a tie, the tie-breaking criteria specified in the rulebook will be applied.
- Results will be announced by the organisers after the final evaluation.
- The decision of the judges and organising committee will be final and binding.

9. Prizes and Certificates

- **Prize Pool:** ₹13k+ (including goodies)
- Certificates will be provided as determined by the organisers.
- Prizes are awarded only to participants who have complied with all rules of the event.

10. Disputes and Modifications

- In case of any dispute or discrepancy, the decision of the organisers and judges shall be final and binding.
- The organisers reserve the right to modify any or all of these rules. Any changes will be communicated to registered participants.

Contact

Event Coordinator: Anirudh Mishra

Phone: 9140463181

Scan to join WhatsApp group